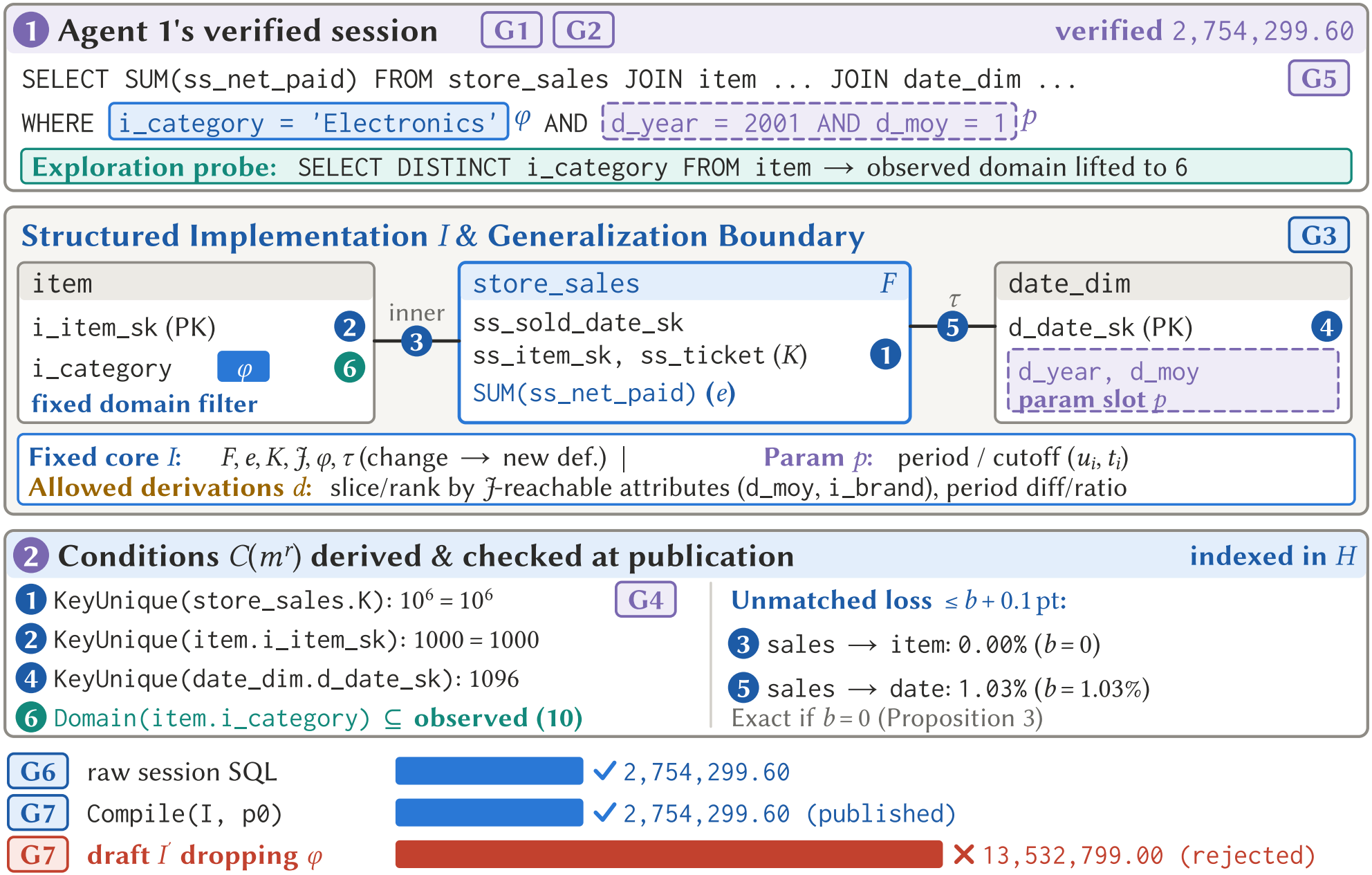

1 Agent 1's verified session
G1
G2
verified 2,754,299.60
SELECT SUM(ss_net_paid) FROM store_sales JOIN item ... JOIN date_dim ...
G5
WHERE i_category = 'Electronics' φ AND d_year = 2001 AND d_moy = 1 p
Exploration probe: SELECT DISTINCT i_category FROM item → observed domain lifted to 6
Structured Implementation I & Generalization Boundary
G3
item
i_item_sk (PK)
i_category
φ
fixed domain filter
inner
store_sales
F
ss_sold_date_sk
ss_item_sk, ss_ticket (K)
SUM(ss_net_paid) (e)
τ
date_dim
d_date_sk (PK)
d_year, d_moy
param slot p
Fixed core I: F, e, K, J, φ, τ (change → new def.) | Param p: period / cutoff (u_i, t_i)
Allowed derivations d: slice/rank by J-reachable attributes (d_moy, i_brand), period diff/ratio
2 Conditions C(m^r) derived & checked at publication
indexed in H
1 KeyUnique(store_sales.K): 10^6 = 10^6
2 KeyUnique(item.i_item_sk): 1000 = 1000
4 KeyUnique(date_dim.d_date_sk): 1096
6 Domain(item.i_category) ⊆ observed (10)
G4
Unmatched loss ≤ b + 0.1 pt:
3 sales → item: 0.00% (b = 0)
5 sales → date: 1.03% (b = 1.03%)
Exact if b = 0 (Proposition 3)
G6 raw session SQL
✓ 2,754,299.60
G7 Compile(I, p0)
✓ 2,754,299.60 (published)
G7 draft I′ dropping φ
✗ 13,532,799.00 (rejected)